COURSE CODE: CS-114

COURSE NAME: Fundamentals of Computer Programming

CREDIT HOURS:

Theory	= 2
Practical	= 1
Total	= 3

CONTACT HOURS:

Theory	= 32
Practical	= 48
Total	= 80

PREREQUISITE: None

MODE OF TEACHING:

Instruction:	Two hours of lecture per week	67%
Practical/ Laboratory Demonstration:	Three hours of Lab work per week	33%

COURSE DESCRIPTION:

This course focuses on the development of fundamental computer concepts and its application to the real world. A systematic approach is used to teach students the basic computer components along with teaching them how to write computer programs that solve well specified problems. Emphasis is placed on the mastery of basic programming skills, with a considerable attention to the fundamental building blocks of computer programs, and the associated concepts and principles. The essentials of sequential processing and control flow are taught in a procedural programming context prior to introducing classes, objects and related object-oriented programming concepts. To ensure the development of the necessary competencies, assigned homework includes the development of program solutions to problems of adequate complexity and relevance.

COURSE OBJECTIVES:

The learning objectives are:

1. Developing comprehensive knowledge about the fundamental principles, concepts and constructs of computer systems and modern computer programming.

2. Developing competencies for the design, coding and debugging of computer programs.

RELEVANT PROGRAM LEARNING OUTCOMES (PLOs):

The course is designed so that students will achieve the PLOs:

1	Engineering Knowledge:	☐	7	Environment and Sustainability:	☐
2	Problem Analysis:	☐	8	Ethics:	☐
3	Design/Development of Solutions:	☐	9	Individual and Team Work:	☐
4	Investigation:	☐	10	Communication:	☐
5	Modern Tool Usage:	☑	11	Project Management:	☐
6	The Engineer and Society:	☐	12	Lifelong Learning:	☐

COURSE LEARNING OUTCOMES:

1. Upon successful completion of the course, the students will be able to:

S.No	CLO	Domain	Taxonomy Level	PLO
1.	Comprehend the fundamental concepts of computer systems.	Cognitive	2	
2.	Apply programming skills to implement computer-based solutions of well specified problems.	Cognitive	3	5
3.	Analyze where computer programs fit in the provision of computer-based solutions to real world problems.	Cognitive	4	5

TOPICS COVERED:

Lecture Breakdown:	
Week	**Topics**
1.	Introduction to Course, Introduction to Computers: System components, networks and operating systems Lab: Computer Hardware and Operating System
2.	Introduction to Logic, Flow Charts, Pseudo Code, Introduction to Programming Lab: Problem Solving using Flow Charts (Visual Logic), Testing the Programming Environment and Basic Program
3.	Programming Environment, Basic Programming, Statements, Expressions, Comments Lab: First Program, Basic statement, and Output Formats
4.	First Program, Compilation, Variables, Types, Operators, Debugging Lab: Variable processing, Arithmetic operations, Finding Errors
5.	I/O, interactive programming Lab: User Input based programming
6.	Relational operators, logical operators, conditional statements Lab: Conditional Constructs I
7.	If-Else Statement, Switch-Break-Continue Lab: Conditional Constructs II
8.	Iterations, For loop, Do-While Loop, While Loop Lab: Loops and Iterations
9.	Functions, parameters to functions Lab: Function Definition, Function Parameters and calls
10.	More functions, Recursion Lab: Function reuse and recursion
11.	Arrays Lab: Arrays and Operations on Arrays
12.	Structures Lab: Structure definition and usage
13.	Pointers Lab: Pointers, Variable Pointer

14.	File I/O Lab: File Handling and Information storage
15.	Project demos, Revision Lab: Project Demos

TEXT AND MATERIAL:

Books:	
Text Book:	1. Peter Norton, Introduction to Computers, McGraw-Hill, 7th Edition, 2013. 2. Paul J. Deitel and Harvey M. Deitel, C++: How to Program, Prentice Hall, 8th/9th Ed.
Reference Books:	• Herbert Schildt, C++: The Complete Reference, 4th Ed. • Robert W. Sebesta, Concepts of Programming Languages, Pearson Education India, 1993. • Noel Kalicharan, C by Example, Cambridge University Press, 1994.

ASSESMENT SYSTEM:

Assessments
Theory: 67%
• Quizzes: 15%
• Assignments: 10%
• OHT-1: 15%
• OHT-2: 15%
• End Semester Exam: 45%
Practical: 33%
• Labs Assignments: 70%
• Semester Project: 30%

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